



Department of **Computing And Information System**

Object Oriented Programming Lab Report-1

Lab Report Name: Java Basics

Course Name: Object Oriented Programming Lab

Course Code: CIS216L

Submitted By:

MD. Shahriar Sanjid Snikdho

Reg ID : 253-16-021

Semester: Fall 2026

Submitted To:

Mr. Md. Nasimul Kader

Assistant Professor

Department of Computing and Information System

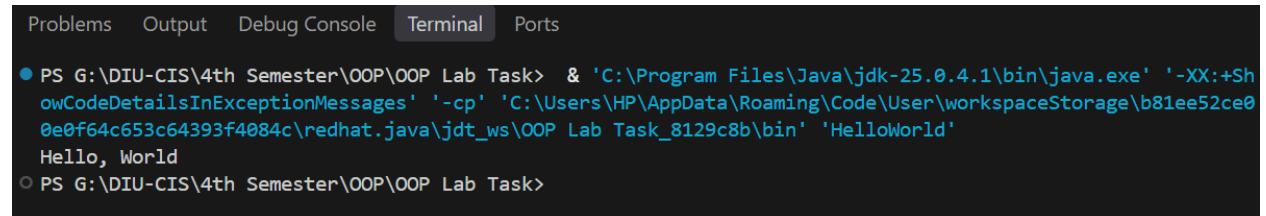
Date of Submission: 26.09.2026

GitHub Repository Link: https://github.com/shahriarsnikdho/OOP_LAB_TASK

Hello World

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello, World");  
    }  
}
```

Output:



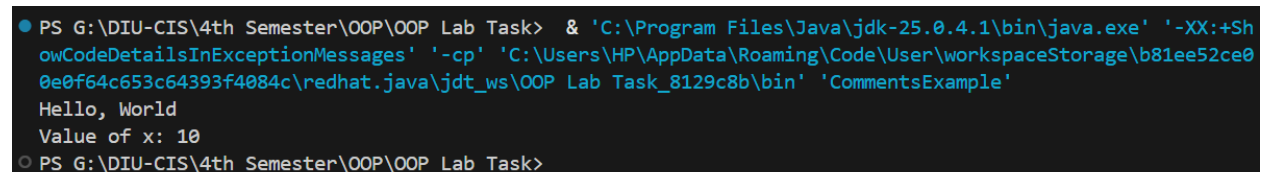
The screenshot shows a terminal window with the following content:

```
Problems Output Debug Console Terminal Ports  
● PS G:\DIU-CIS\4th Semester\OOP\OOP Lab Task> & 'C:\Program Files\Java\jdk-25.0.4.1\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP\AppData\Roaming\Code\User\workspaceStorage\b81ee52ce00e0f64c653c64393f4084c\redhat.java\jdt_ws\OOP Lab Task_8129c8b\bin' 'HelloWorld'  
Hello, World  
○ PS G:\DIU-CIS\4th Semester\OOP\OOP Lab Task>
```

Comments Example

```
public class CommentsExample {  
    public static void main(String[] args) {  
        // This is a single-line comment  
        System.out.println("Hello, World"); // Comment at end of line  
  
        /*  
        * This is a multi-line comment  
        */  
        int x = 10;  
        System.out.println("Value of x: " + x);  
    }  
}
```

Output:



The screenshot shows a terminal window with the following content:

```
● PS G:\DIU-CIS\4th Semester\OOP\OOP Lab Task> & 'C:\Program Files\Java\jdk-25.0.4.1\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP\AppData\Roaming\Code\User\workspaceStorage\b81ee52ce00e0f64c653c64393f4084c\redhat.java\jdt_ws\OOP Lab Task_8129c8b\bin' 'CommentsExample'  
Hello, World  
Value of x: 10  
○ PS G:\DIU-CIS\4th Semester\OOP\OOP Lab Task>
```

Datatypes Demo

```
public class DataTypesDemo {  
    public static void main(String[] args) {  
  
        // ----- Primitive Data Types -----  
        byte b = 100; // 1 byte
```

```

short s = 30000; // 2 bytes
int i = 100000; // 4 bytes
long l = 10000000000L; // 8 bytes

float f = 3.14f; // 4 bytes
double d = 3.14159265359; // 8 bytes

char c = 'A'; // 2 bytes (Unicode character)
boolean flag = true; // 1 bit

// ----- Non-Primitive Data Types -----
String str = "Hello, Java"; // String (class in Java)
int[] arr = {1, 2, 3, 4, 5}; // Array
Integer wrapperInt = Integer.valueOf(42); // Wrapper class
StringBuilder sb = new StringBuilder("Java"); // Class object

// ----- Output -----
System.out.println("byte: " + b);
System.out.println("short: " + s);
System.out.println("int: " + i);
System.out.println("long: " + l);
System.out.println("float: " + f);
System.out.println("double: " + d);
System.out.println("char: " + c);
System.out.println("boolean: " + flag);

System.out.println("String: " + str);
System.out.print("Array: ");
for (int num : arr) {
    System.out.print(num + " ");
}
System.out.println();

System.out.println("Wrapper Integer: " + wrapperInt);
System.out.println("StringBuilder: " + sb);
}
}

```

Output:

```
Problems 1 Output Debug Console Terminal Ports
• PS G:\DIU-CIS\4th Semester\OOP\OOP Lab Task> & 'C:\Program Files\Java\jdk-25.0.4.1\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP\AppData\Roaming\Code\User\workspaceStorage\b81ee52ce00e0f64c653c64393f4084c\redhat.java\jdt_ws\OOP Lab Task_8129c8b\bin' 'DataTypesDemo'
byte: 100
short: 30000
int: 100000
long: 10000000000
float: 3.14
double: 3.14159265359
char: A
boolean: true
String: Hello, Java
Array: 1 2 3 4 5
Wrapper Integer: 42
StringBuilder: Java
○ PS G:\DIU-CIS\4th Semester\OOP\OOP Lab Task>
```

Variables Demo

```
public class VariablesDemo {

    // Instance variable (belongs to each object)
    int instanceVar = 10;

    // Static variable (shared across all objects of the class)
    static String staticVar = "I am static";

    public void showVariables() {
        // Local variable (declared inside a method)
        int localVar = 5;

        System.out.println("Instance Variable: " + instanceVar);
        System.out.println("Static Variable: " + staticVar);
        System.out.println("Local Variable: " + localVar);
    }

    public static void main(String[] args) {
        // Creating object
        VariablesDemo obj1 = new VariablesDemo();
        obj1.showVariables();

        // Accessing static variable directly using class name
        System.out.println("Accessing Static Variable via class: " +
VariablesDemo.staticVar);
    }
}
```

Output:

```

● PS G:\DIU-CIS\4th Semester\OOP\OOP Lab Task> & 'C:\Program Files\Java\jdk-25.0.4.1\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP\AppData\Roaming\Code\User\workspaceStorage\b81ee52ce00e0f64c653c64393f4084c\redhat.java\jdt_ws\OOP Lab Task_8129c8b\bin' 'VariablesDemo'
Instance Variable: 10
Static Variable: I am static
Local Variable: 5
Accessing Static Variable via class: I am static
○ PS G:\DIU-CIS\4th Semester\OOP\OOP Lab Task>

```

Simple Operators Demo

```

public class SimpleOperatorsDemo {
    public static void main(String[] args) {
        int a = 10, b = 3;

        // Arithmetic Operators
        System.out.println("a + b = " + (a + b)); // Addition
        System.out.println("a - b = " + (a - b)); // Subtraction

        // Relational Operator
        System.out.println("a > b ? " + (a > b)); // Greater than

        // Logical Operator
        boolean x = true, y = false;
        System.out.println("x && y = " + (x && y)); // Logical AND

        // Assignment Operator
        a += 5; // a = a + 5
        System.out.println("a after += 5 : " + a);

        // Ternary Operator
        int max = (a > b) ? a : b;
        System.out.println("Maximum = " + max);
    }
}

```

Output:

```

● PS G:\DIU-CIS\4th Semester\OOP\OOP Lab Task> & 'C:\Program Files\Java\jdk-25.0.4.1\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP\AppData\Roaming\Code\User\workspaceStorage\b81ee52ce00e0f64c653c64393f4084c\redhat.java\jdt_ws\OOP Lab Task_8129c8b\bin' 'SimpleOperatorsDemo'
a + b = 13
a - b = 7
a > b ? true
x && y = false
a after += 5 : 15
Maximum = 15
○ PS G:\DIU-CIS\4th Semester\OOP\OOP Lab Task>

```

Decision Making Demo

```

public class DecisionMakingDemo {
    public static void main(String[] args) {

```

```

int number = 10;

// if statement
if (number > 0) {
    System.out.println("The number is positive.");
}

// if-else statement
if (number % 2 == 0) {
    System.out.println("The number is even.");
} else {
    System.out.println("The number is odd.");
}

// if-else-if ladder
if (number < 0) {
    System.out.println("The number is negative.");
} else if (number == 0) {
    System.out.println("The number is zero.");
} else {
    System.out.println("The number is positive.");
}

// switch statement
int day = 3;
switch (day) {
    case 1:
        System.out.println("Monday");
        break;
    case 2:
        System.out.println("Tuesday");
        break;
    case 3:
        System.out.println("Wednesday");
        break;
    default:
        System.out.println("Other day");
}
}

```

Output:

```
● PS G:\DIU-CIS\4th Semester\OOP\OOP Lab Task> & 'C:\Program Files\Java\jdk-25.0.4.1\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP\AppData\Roaming\Code\User\workspaceStorage\b81ee52ce00e0f64c653c64393f4084c\redhat.java\jdt_ws\OOP Lab Task_8129c8b\bin' 'DecisionMakingDemo'
The number is positive.
The number is even.
The number is positive.
Wednesday
```

Loops Demo

```
public class LoopsDemo {
    public static void main(String[] args) {

        // 1. For loop
        System.out.println("For Loop:");
        for (int i = 1; i <= 5; i++) {
            System.out.println("i = " + i);
        }

        // 2. While loop
        System.out.println("\nWhile Loop:");
        int j = 1;
        while (j <= 5) {
            System.out.println("j = " + j);
            j++;
        }

        // 3. Do-While loop
        System.out.println("\nDo-While Loop:");
        int k = 1;
        do {
            System.out.println("k = " + k);
            k++;
        } while (k <= 5);

        // 4. Enhanced For Loop (for-each loop)
        System.out.println("\nEnhanced For Loop:");
        int[] numbers = {10, 20, 30, 40, 50};
        for (int num : numbers) {
            System.out.println("num = " + num);
        }
    }
}
```

Output:

```
PS G:\DIU-CIS\4th Semester\OOP\OOP Lab Task> & 'C:\Program Files\Java\jdk-25.0.4.1\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP\AppData\Roaming\Code\User\workspaceStorage\b81ee52ce00e0f64c653c64393f4084c\redhat.java\jdt_ws\OOP Lab Task_8129c8b\bin' 'LoopsDemo'
```

```
k = 1  
k = 2  
k = 3  
k = 4  
k = 5
```

Enhanced For Loop:

```
num = 10  
num = 20  
num = 30  
num = 40  
num = 50
```